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ORIGINAL ARTICLE

Leptospira Genomospecies and Sequence Type Prevalence in Small Mammal Populations in Germany

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Abstract

Leptospirosis is a worldwide emerging infectious disease caused by zoonotic bacteria of the genus *Leptospira*. Numerous mammals, including domestic and companion animals, can be infected by *Leptospira* spp., but rodents and other small mammals are considered the main reservoir. The annual number of recorded human leptospirosis cases in Germany (2001–2016) was 25–166. Field fever outbreaks in strawberry pickers, due to infection with *Leptospira kirschneri* serovar Grippotyphosa, were reported in 2007 and 2014. To identify the most commonly occurring *Leptospira* genomospecies, sequence types (STs), and their small mammal host specificity, a monitoring study was performed during 2010–2014 in four federal states of Germany. Initial screening of kidney tissues of 3,950 animals by PCR targeting the *lipl32* gene revealed 435 rodents of 6 species and 89 shrews of three species positive for leptospiral DNA. PCR-based analyses resulted in the identification of the genomospecies *L. kirschneri* (62.7%), *Leptospira interrogans* (28.3%), and *Leptospira borgpetersenii* (9.0%), which are represented by four, one, and two STs, respectively. The average *Leptospira* prevalence was highest ($\sim$30%) in common voles (*Microtus arvalis*) and field voles (*Microtus agrestis*). Both species were exclusively infected with *L. kirschneri*. In contrast, in bank voles (*Myodes glareolus*) and yellow-necked mice (*Apodemus flavicollis*), DNA of all three genomospecies was detected, and in common shrews (*Sorex araneus*) DNA of *L. kirschneri* and *L. borgpetersenii* was identified. The association between individual infection status and demographic factors varied between species; infection status was always positively correlated to body weight. In conclusion, the study confirmed a broad geographical distribution of *Leptospira* in small mammals and suggested an important public health relevance of common and field voles as reservoirs of *L. kirschneri*. Furthermore, the investigations identified seasonal, habitat-related, as well as individual influences on *Leptospira* prevalence in small mammals that might impact public health.

Keywords: demography, Germany, habitat, *Leptospira* spp., leptospirosis, MLST, rodent, shrew, SLST

Introduction

Leptospirosis is a widespread and potentially fatal zoonosis and is also a major cause of disease in both domestic and companion animals (Ayral et al. 2014). Currently, 22 *Leptospira* genomospecies are described, subdivided into a pathogenic, a nonpathogenic, and an intermediate group (Fouts et al. 2016). Serologically, the genomospecies are subdivided into more than 300 serovars, which, if antigenically related, are grouped into serogroups (Levett 2001, 2015, Mayer-Scholl et al. 2014). Severity of human disease varies with host susceptibility and serovar (Lessa-Aquino et al. 2017). Most human *Leptospira* infections are mild and resolve spontaneously (Levett 2001, Victoriano et al. 2009). Severe leptospirosis is characterized by sudden onset of fever, chills, and headache and can progress to dysfunction of multiple organs such as kidney, liver, lungs, and brain (Faine 1999, Levett 2001, Watt 2010). The mortality in humans can reach 15% when the disease is untreated (Ko et al. 1999, Yanagihara et al. 2007, Meerburg et al. 2009).

Exposure may occur either through direct contact with infected animals or indirectly through contact with water or

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soil contaminated with urine from infected animals (Levett 2001, Watt 2010). Major leptospirosis outbreaks usually occur in tropical and subtropical regions. In contrast, in temperate areas, *Leptospira* usually may cause sporadic epidemics (Watt 2010). In Germany, the annual human incidence ranged during 2001–2016 from 0.06 to 0.22 per 100,000 inhabitants (Robert Koch-Institute 2016). The incidence increased in 2007 and 2014 due to disease clusters among strawberry pickers in North Rhine-Westphalia and Lower Saxony, respectively (Desai et al. 2009, Dreesman et al. 2016).

Worldwide, many epidemiological surveys have shown that rodents and other small mammals are the most common reservoirs for leptospires (Gamage et al. 2011, Perez et al. 2011, Calderon et al. 2014, Andersen-Ranberg et al. 2016). In Germany, studies have shown leptospires not only in wild and commensal rodents of different species, including voles, beavers, mice, and rats, but also in shrews (Mochmann 1957, Kocianova et al. 1993, Woll et al. 2012, Runge et al. 2013, Obiegala et al. 2016, 2017, Heuser et al. 2017).

Reasons for the occurrence of human leptospirosis clusters or outbreaks in temperate areas, especially in industrialized countries like Germany, remain to be investigated, because their knowledge is essential for the development of early-warning systems and the application of appropriate public health measures. Therefore, this study aimed on evaluating the small mammal host specificity regarding *Leptospira* genomospecies and their sequence types (STs). Furthermore, the relationship between infection status and host demography as well as geographical, seasonal, and habitat-related influences on *Leptospira* prevalence was studied.

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Materials and Methods

Rodent trapping and sample collection

Small mammals were trapped in spring, summer, and autumn from 2010 to 2014 in forest and grassland habitats at four locations in Germany (Reil et al. 2017) (Fig. 1). Per site and habitat, three plots were selected by conducting live and snap trapping, the latter according to the standard protocol of federal forest institutions (Reil et al. 2017). All animals found dead in live traps and from snap trapping were included in the study. Individuals were frozen at −20°C until dissection according to in-house standard protocols. During dissection, the entire kidney was collected and transferred into a 1.5 mL tube and stored at −20°C until further examination. Furthermore, tail and other tissue samples were collected and species, weight, sex, and reproductive status of the animals were recorded.

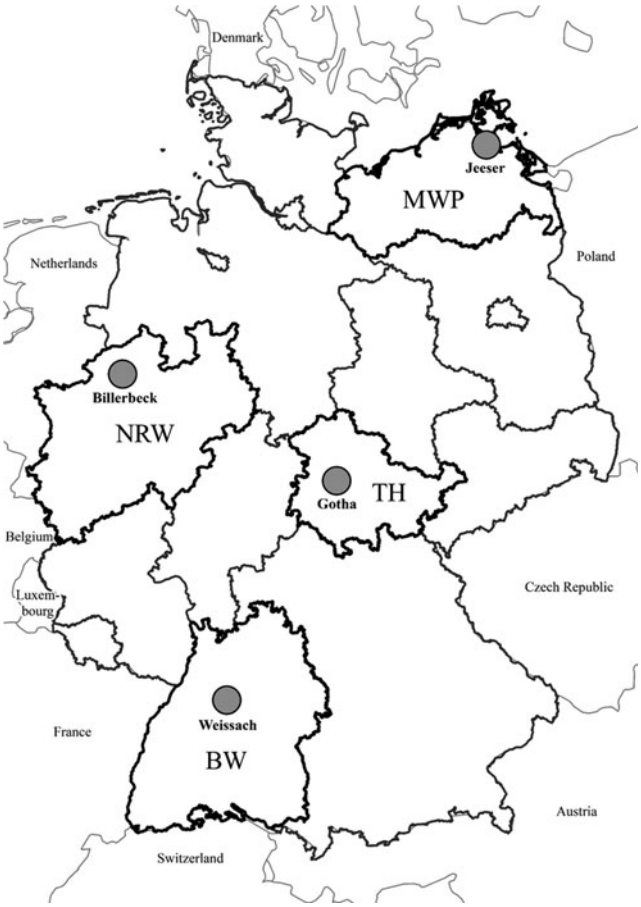


FIG. 1. Location of the four trapping sites in Germany. Snap and live trapping were performed in spring, summer, and autumn 2010–2014 on three replicate plots per forest and grassland habitat. The plots were located near Jeeser (coordinates: latitude 54.160620, longitude 13.259261; district of Vorpommern-Rügen; MWP), Billerbeck (coordinates: latitude 51.988741, longitude 7.323869; district of Coesfeld; NRW), Gotha (coordinates: latitude 50.971375, longitude 10.653304; district of Gotha; TH), and Weissach (coordinates: latitude 48.833471, longitude 8.958623; district of Böblingen; BW). BW, Baden-Wuerttemberg; MWP, Mecklenburg-Western Pomerania; NRW, North Rhine-Westphalia; TH, Thuringia. © Map of Germany: Federal Agency for Cartography and Geodesy.

Detection and typing of Leptospira *DNA*

DNA was isolated from a small piece of the kidney tissue sample using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany). The initial *lipl32* PCR was used to detect *Leptospira* DNA in the kidney-derived DNA preparations (Mayer-Scholl et al. 2011). PCR-positive samples were further analyzed by single locus sequence typing (SLST) and multilocus sequence typing (MLST) (Victoria et al. 2008, Boonsilp et al. 2013).

Molecular species and sex determination of small mammals

For molecular confirmation of small mammal species and sex determination, DNA was isolated from tail or other tissues using the GeneMATRIX Tissue DNA Purification Kit (Roboklon, Berlin, Germany) or a tail tissue lysis protocol (Schmidt et al. 2016). PCR-based cytochrome *b* analysis was used to confirm the species of *Leptospira*-positive small mammals and animals where the morphological determination was impossible because of a poor condition of the carcass (Schlegel et al. 2012). If sex determination was necessary, multiplex PCR approaches targeting the *ZFX*, *ZFY*, and *Sry* genes and the *DBY1*, *DBY3*, *DBY7*, *UTY11*, and *Sry* genes were applied for rodents and shrews, respectively (Aasen and Medrano 1990, Bryja and Konecny 2003, Yannic et al. 2008).

Statistics

Analyses were designed to quantify factors related to individual infection status in different small mammal species.

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Individual infection status (positive or negative) was analyzed using a generalized linear mixed model (GLMM) approach for three objectives: first, the association between host species and *Leptospira* spp. prevalence was investigated, including the spatial design (federal state, habitat) of the study. Second, the influence of season and geographic region (federal state) in combination with demographic factors on the individual infection status was characterized. Third, seasonal change in *Leptospira* spp. prevalence and habitat effects on the combined small mammal community were assessed. In the models, binomial error distribution and a logit link function were used with random factors accounting for the nested spatial design of the study (site nested in federal state). All analyses were performed in R (version 3.3.2; R Core Team 2015). Level of significance was $\alpha < 5\%$.

To determine host specificity, the individual infection status with *Leptospira* spp. and one of the genomospecies was used as the binomial dependent variable and modeled in relationship to the respective host species. To estimate species-specific infection probabilities, least square means were obtained from the *lsmeans* package to back transform estimates from logit scale based on the reference GLMM, resulting in infection probabilities that were then used to visualize host specificity. Only species with more than 10 trapped individuals were analyzed.

To analyze factors modulating the individual infection status, species-specific global models were constructed using individual demographic factors (sex, reproductive activity, and weight) and their respective two-way interactions, as well as the corresponding trapping season and federal state. Backward model simplification was performed using likelihood ratio tests. If season or federal state remained in the final model, within-subject contrasts were compared using pairwise ''Tukey'' comparisons of all levels using the *glht* function from the *multcomp* package.

On the level of the whole small mammal community, the variation in seasonal population dynamics might result in overall differences in *Leptospira* spp. prevalence between the two investigated habitat types (grassland versus forest). The *Leptospira* spp. prevalence data from all species were combined per habitat and compared for each season.

Results

Collection of small mammals

A total of 4,023 rodents and other small mammals were collected at the 4 trapping sites in Germany from 2010 to 2014. Overall, 3,306 rodents of 4 murine species (family Muridae) and 4 vole species (family Cricetidae), 715 insectivores (families Soricidae and Talpidae), and 2 least weasels (*Mustela nivalis*) were collected Table 1. For 3,950 individuals, kidney tissue was available for subsequent *Leptospira* PCR analysis (Table 1).

Prevalence of Leptospira *infection, genomospecies, and ST determination*

Using the screening *lipl32* PCR, 524 of 3,950 (13.3%) small mammals were tested positive for *Leptospira*-specific DNA. *Leptospira* DNA was detected in six rodent and all three *Sorex* shrew species (Table 2). Common and field voles were most often infected by *Leptospira* spp. (30.0% and

TABLE 1. NUMBER AND SPECIES OF SMALL MAMMALS COLLECTED DURING 2010–2014 IN FOUR FEDERAL STATES OF GERMANY AND KIDNEY SAMPLES OBTAINED

				Total no. of animals/kidney samples	*MWP*		*NRW*		*TH*		*BW*	
Order	*Family*	*Common name*	*Scientific name*		*Total*	*investigated*	*Total*	*investigated*	*Total*	*investigated*	*Total*	*investigated*
Rodentia	Muridae	Striped field mouse	*Apodemus agrarius*	45/44	36	35	0	0	9	9	0	0
		Yellow-necked mouse	*Apodemus flavicollis*	930/925	219	219	139	139	297	296	275	271
		Wood mouse	*Apodemus sylvaticus*	105/105	0	0	58	58	16	16	31	31
		Eurasian harvest mouse	*Micromys minutus*	6/6	2	2	4	4	0	0	0	0
	Cricetidae	Bank vole	*Myodes glareolus*	1,612/1,578	268	266	336	324	493	486	515	502
		Common vole	*Microtus arvalis*	395/377	83	80	20	20	127	121	165	156
		Field vole	*Microtus agrestis*	209/209	23	23	1	1	178	178	7	7
		European water vole	*Arvicola* spp.	4/4	0	0	1	1	3	3	0	0
Eulipotyphla	Soricidae	Common shrew	*Sorex araneus*	530/521	232	226	20	20	254	251	24	24
		Crowned shrew	*Sorex coronatus*	102/101	11	11	33	32	0	0	58	58
		Eurasian pygmy shrew	*Sorex minutus*	77/73	20	19	9	8	17	16	31	30
		Eurasian water shrew	*Neomys fodiens*	4/4	4	4	0	0	0	0	0	0
		Greater white-toothed shrew	*Crocidura russula*	1/1	0	0	1	1	0	0	0	0
	Talpidae	European mole	*Talpa europaea*	1/0	0	0	0	0	1	0	0	0
Carnivora	Mustelidae	Least weasel	*Mustela nivalis*	2/2	1	1	0	0	1	1	0	0
Total				4,023/3,950	899	886	622	608	1,396	1,377	1,106	1,079

BW, Baden-Wuerttemberg; MWP, Mecklenburg-Western Pomerania; NRW, North Rhine-Westphalia; TH, Thuringia.

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TABLE 2. DETECTION OF *LEPTOSPIRA* SPP. DNA IN RODENT AND SHREW SPECIES IN GERMANY THROUGH *LIPL32* PCR

	No. of Leptospira*-positive/total no. of investigated animals (percentage of positive samples) for mouse, vole, and shrew species*										
		Mouse				*Vole*			*Shrew*		
Federal state	*Total*	*Striped field*	*Yellow-necked*	*Wood*	*Field*	*Common*	*Bank*	*Common*	*Crowned*	*Eurasian pygmy*
MWP	150/886 (16.9%)	5/35 (14.3%)	31/219 (14.2%)	—	4/23 (17.4%)	37/80 (46.3%)	16/266 (6.0%)	53/226 (23.5%)	0/11 (0.0%)	4/19 (21.1%)
NRW	129/608 (21.2%)	—	50/139 (36.0%)	8/58 (13.8%)	0/1 (0.0%)	2/20 (10.0%)	66/324 (20.4%)	1/20 (5.0%)	1/32 (3.1%)	1/8 (12.5%)
TH	170/1,377 (12.3%)	1/9 (11.1%)	27/296 (9.1%)	1/16 (6.3%)	58/178 (32.6%)	28/121 (23.1%)	28/486 (5.8%)	27/251 (10.8%)	—	0/16 (0.0%)
BW	75/1,079 (7.0%)	—	14/271 (5.2%)	0/31 (0.0%)	0/7 (0.0%)	46/156 (29.5%)	13/502 (2.6%)	0/24 (0.0%)	2/58 (3.4%)	0/30 (0.0%)
Total	524/3,950 (13.3%)	6/44 (13.6%)	122/925 (13.2%)	9/105 (8.6%)	62/209 (29.7%)	113/377 (30.0%)	123/1,578 (7.8%)	81/521 (15.5%)	3/101 (3.0%)	5/73 (6.8%)

—, No animal of this species trapped.

29.7%), followed by common shrews (15.5%) and yellow-necked mice (13.2%) (Table 2 and Fig. 2A).

The partial *secY* gene-based SLST was able to successfully identify the presence of *Leptospira* DNA in 279 (53%) samples. The genomospecies identified were *Leptospira kirschneri* (62.7%), *Leptospira interrogans* (28.3%), and *Leptospira borgpetersenii* (9.0%) (Fig. 2B and Supplementary Fig. S1A; Supplementary Data are available online at www.liebertpub.com/vbz).

The subsequent MLST resulted in the identification of 7 STs in 152 of 279 (55%) samples (Fig. 3). *L. kirschneri* was represented by ST 110, ST 117, ST 136, and ST 230. *L. borgpetersenii* included ST 146 and ST 197, whereas *L. interrogans* was exclusively represented by ST 24. The serovars corresponding to the detected STs are shown in Table 3.

Host specificity of Leptospira *genomospecies and STs*

The association of *Leptospira* genomospecies with small mammals showed multiple patterns: one to three *Leptospira* genomospecies were detected per host species (Supplementary Fig. S1A). Common and field voles were more frequently infected by *L. kirschneri* than any other small mammal species (Fig. 2B). Similarly, striped field mouse, wood mouse, and crowned shrew were infected by *L. kirschneri*, *L. interrogans*, or *L. borgpetersenii*. In contrast, bank voles, yellow-necked mice, and common shrews were infected with multiple *Leptospira* species (Fig. 2B and Supplementary Fig. S1A).

The subsequent MLST demonstrated that for five of eight small mammal species, each genomospecies was represented by a single ST (Fig. 3). *L. kirschneri* in field voles and yellow-necked mice was exclusively represented by ST 110 and in striped field mice by ST 117. In common voles, *L. kirschneri* was also mainly represented by ST 110, but ST 230 emerged in the common vole population in Baden-Wuerttemberg (Supplementary Fig. S2D). *L. kirschneri* in common shrews was represented by ST 110 and ST 136 and *L. borgpetersenii* in bank voles by ST 146 and ST 197 (Fig. 3). *L. interrogans* was represented in yellow-necked mice, wood mice, and bank voles exclusively by ST 24.

Geographical distribution and seasonal pattern of Leptospira *infections*

The average percentage of *Leptospira*-positive animals in the federal states ranged between 7.0% and 21.2% (Table 2). The individual infection probability in common voles, yellow-necked mice, and bank voles differed significantly among some federal states ($p \leq 0.001$–0.028) (Table 4). Common and field voles were the main carriers of *Leptospira* spp. throughout the trapping sites with the highest prevalence in autumn ($p < 0.001$ and $p = 0.006$, respectively; Table 4, Supplementary Fig. S3A, B). Leptospiral DNA was also found in yellow-necked mice and bank voles in all regions. Similarly, striped field mice were infected with *Leptospira* spp. in both regions they could be trapped. PCR-positive shrews were detected in all four regions with no differences between federal states (Table 4).

All three *Leptospira* genomospecies were detected in each federal state (Supplementary Fig. S1B–E). In common shrews (from two of four regions) and yellow-necked mice

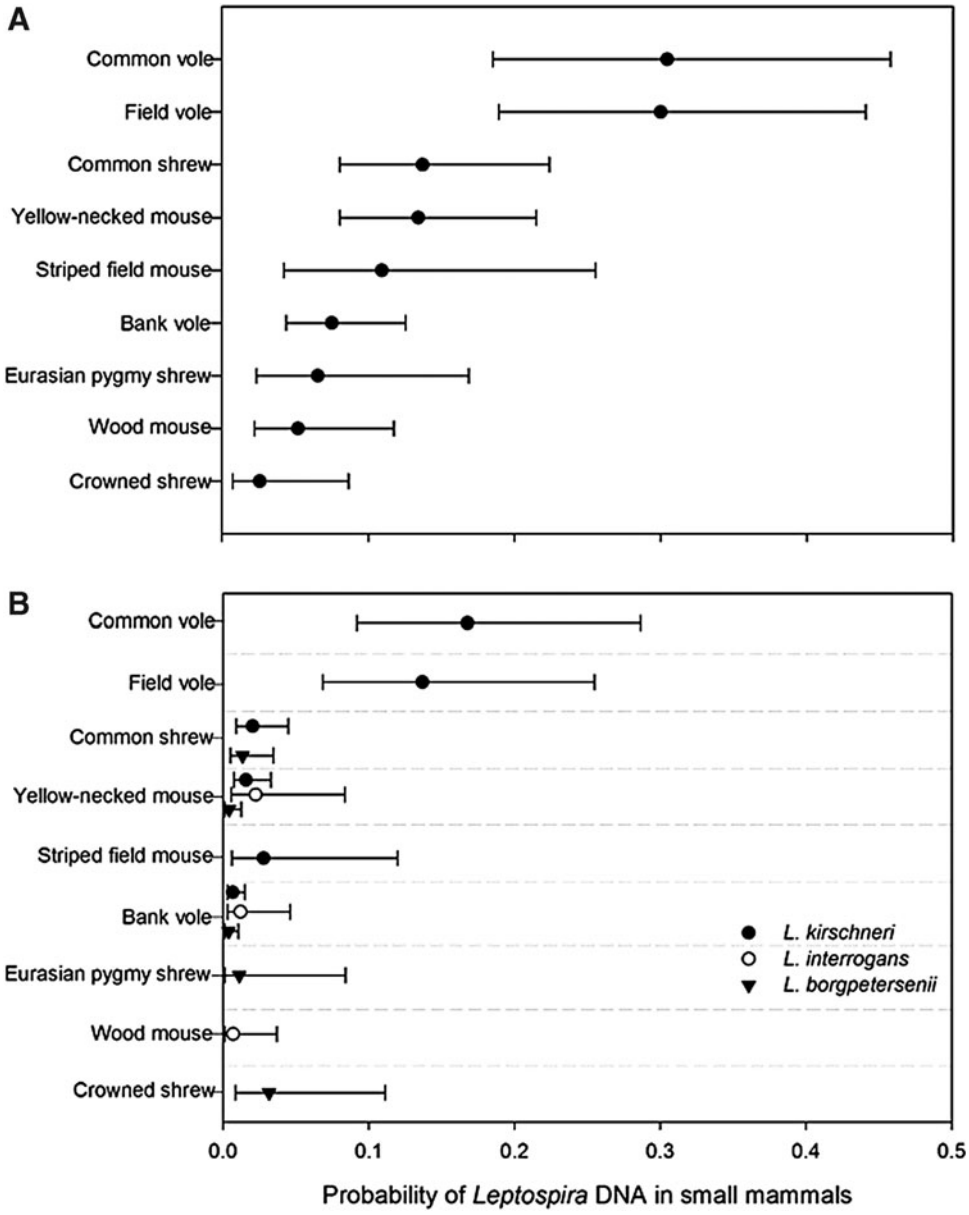


FIG. 2. Probability of *Leptospira* DNA detection **(A)** and detection of DNA of one of the three genomospecies *Leptospira kirschneri*, *Leptospira interrogans*, and *Leptospira borgpetersenii* **(B)**. The *lipl32* PCR resulted in the detection of 515 *Leptospira* DNA-positive small mammals of 9 species **(A)**. The *secY* PCR resulted in the identification of three genomospecies for 274 of 515 *lipl32*-PCR-positive samples **(B)**.

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(from three of four regions), two to three genomospecies were detected (Supplementary Fig. S1B, D, E). ST 24 and ST 110 were detected in animals from all four regions (Supplementary Fig. S2A–D). In contrast, the STs in common shrews differed between the sites in Mecklenburg-Western Pomerania, North Rhine-Westphalia, and Thuringia (Supplementary Fig. S2A–C). The individual probability of *Leptospira* spp. infection in common shrews was higher in spring compared to summer and autumn (Table 4).

Effects of habitat and demography on Leptospira *infection*

The influence of demographic factors on the individual infection status was species specific (Table 4). Body weight (proxy for age) was positively correlated to infection status in all species where demographic factors remained in the final model ($p \leq 0.001$–0.005). The positive estimate of weight indicated that there was a positive association between age and *Leptospira* infection across species. Sex and reproduction status remained in the final model for yellow-necked mice ($p = 0.025$), while heavier males of bank voles had a significantly higher probability of *Leptospira* infection compared to females ($p = 0.019$).

Furthermore, the analysis revealed that throughout the year, individuals caught in open grassland had a higher probability of carrying *Leptospira* DNA compared to individuals caught in forests ($p \leq 0.001$–0.036) (Table 5).

Discussion

Genomospecies and ST association in small mammals

To our knowledge, this is the first study that investigated the *Leptospira* prevalence in small mammals in forest and grassland habitats throughout Germany, taking into account

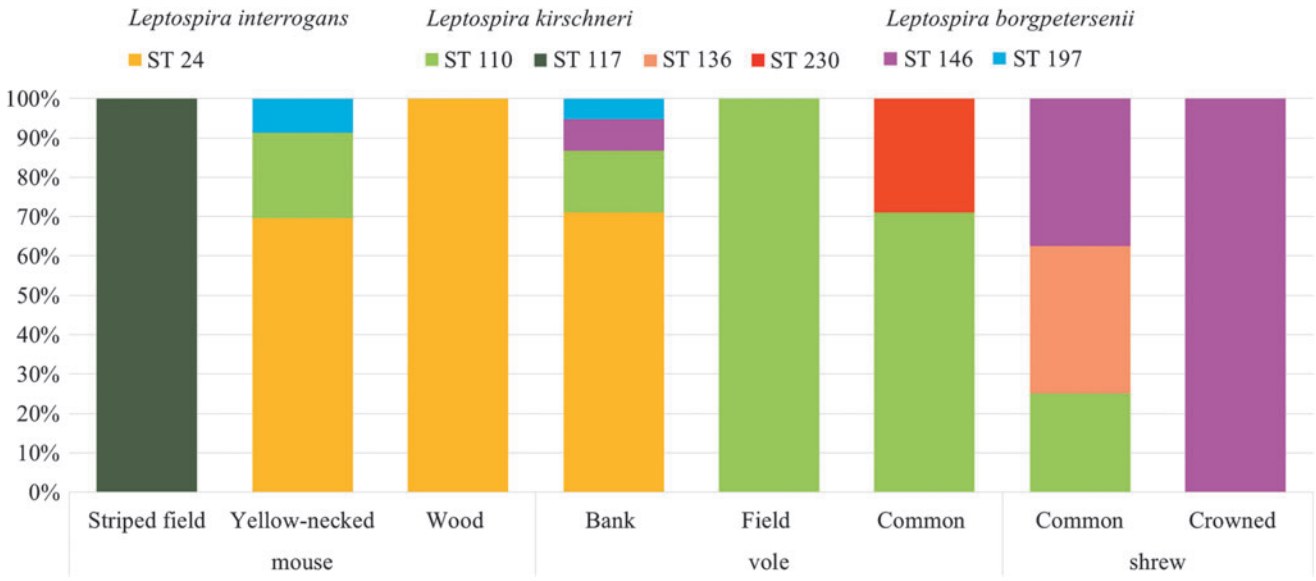


FIG. 3. ST distribution of *L. interrogans* (ST 24), *L. kirschneri* (ST 110, ST 117, ST 136, and ST 230), and *L. borgpetersenii* (ST 146 and ST 197) in small mammal species in Germany as determined by MLST. MLST, multilocus sequence typing; ST, sequence type.

the small mammal host specificity regarding *Leptospira* genomospecies, their respective STs, and the relationship between infection status and host demography, as well as geographical, seasonal, and habitat-related influences on *Leptospira* prevalence over a period of 5 years.

In this study, we confirmed a broad distribution of *Leptospira* DNA in small mammals, such as rodents and shrews in Germany (Mochmann 1957, Woll et al. 2012, Mayer-Scholl et al. 2014, Obiegala et al. 2016, 2017, Heuser et al. 2017). Two patterns of *Leptospira* association were observed. Common and field voles were found in our study with the highest DNA prevalence of ~30% in average (ranging between 10% and 46% across regions), a broad geographical distribution and an exclusive detection of the genomospecies *L. kirschneri*. The overall high *Leptospira* prevalence in common voles also resulted in the generally higher prevalence in open grassland (primary habitat of the common vole) (Jacob et al. 2014) compared to forest. In all field voles and the majority of common voles, ST 110 was exclusively detected, except 13 cases of ST 230 at a single trapping site. These observations are similar to results of a pilot study with a high average prevalence of 13% and the exclusive detection of *L. kirschneri* in common voles (Mayer-Scholl et al. 2014). In contrast to common and field voles, bank voles and yellow-necked mice did not only have a lower average prevalence (8–13%) but also harbored *L. interrogans* ST 24, *L. kirschneri* ST 110, or *L. borgpetersenii* ST 197. A lower prevalence (4–6%) was also observed in bank voles in previous investigations in Germany (Mayer-Scholl et al. 2014, Obiegala et al. 2016, 2017). Furthermore, common shrews were found for the first time to be associated with different STs (110 and 136) of *L. kirschneri* in different federal states. The observed high *Leptospira* spp. prevalence of 15% confirms the findings of Obiegala et al. (2017), suggesting common shrews as additional reservoir.

Table 3. Multilocus Sequence Type and Corresponding Serovars Detected in Common, Field, and Bank Vole, Yellow-Necked, Wood, and Striped Field Mouse, and Common, Crowned, and Eurasian Pygmy Shrew

Sequence type	*Serovar*[a]
24	Bratislava/Jalna
110	Grippotyphosa/Vanderhoedeni/Valbuzzi
117	Mozdok
136	Undesignated
146	Poi/Javanica/Sorexjalna
197	Undesignated
230	Undesignated

[a]http://pubmlst.org MLST scheme No. 1 (Boonsilp et al. 2013).
MLST, multilocus sequence typing.

Pathogen–host association and trans-species transmission

The pattern described above suggests two mechanisms of a potential pathogen–host association. On the one hand, the findings in field vole and common vole samples suggest a strong specificity to *L. kirschneri* ST 110/ST 230 and the habitat grassland, at least for Germany. Similarly, Norway rats were exclusively found to be infected with *L. interrogans* ST 17, which corresponds to serogroup Icterohaemorrhagiae that is responsible for human infections in urban areas (de Faria et al. 2008, Agudelo-Florez et al. 2009, Koizumi et al. 2009, Krojgaard et al. 2009, Scialfa et al. 2010, Theuerkauf et al. 2013, Ayral et al. 2015, Strand et al. 2015, Heuser et al. 2017).

On the other hand, the frequent detection of *L. kirschneri* ST 110 (with a 63% overall prevalence) in yellow-necked mice, bank voles, and common shrews sharing their habitat, may indicate a trans-species transmission. The same is obvious

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Table 4. Effect of Demographic Factors, Season, and State on Individual Infection with *Leptospira* in Common, Field, and Bank Vole, Yellow-Necked, Wood, and Striped Field Mouse, and Common, Crowned, and Eurasian Pygmy Shrew

	Source of variation (fixed effects)														*Random*
Species	*Intercept*	*Sex (m)*	*Reproduction (1)*	*Weight*	*Sex (m): reproduction (1)*	*Sex (m): weight (g)*	*Spring vs. autumn*	*Summer vs. autumn*	*Summer vs. spring*	*MWP vs. BW*	*NRW vs. BW*	*TH vs. BW*	*NRW vs. MWP*	*TH vs. MWP*	*TH vs. NRW*	*State:site*
Common vole (*M. arvalis*, $N=353$)															
Estimate	−3.97			0.074			−1.179	−0.805	0.373	0.774	−1.56	−0.396	−2.334	1.17	1.164	
SE	0.234			0.016			0.678	0.267	0.681	0.308	0.78	0.301	0.797	0.335	0.79	
z-Value	−1.701			4.515			−1.738	−3.02	0.548	2.513	−1.999	−1.31	−2.927	−3.487	1.474	
p	0.089			**<0.001**			0.177	**0.006**	0.839	0.051	0.172	0.532	**0.016**	**0.002**	0.43	
Variation (σ)																0.000
SD																0.000
Field vole (*M. agrestis*, $N=207$)															
Estimate	−0.893			0.091			−2.033	−1.594	0.439							
SE	0.402			0.023			1.255	0.388	1.257							
z-Value	−2.22			4.02			−1.62	−4.11	0.349							
p	**0.026**			**<0.001**			0.217	**<0.001**	0.93							
Variation (σ)																0.421
SD																0.649
Striped field mouse (*A. agrarius*, $N=44$)															
Estimate	−3.178	2.148													
SE	1.021	1.146													
z-Value	−3.114	1.875													
p	**0.002**	0.061													
Variation (σ)																0.000
SD																0.000
Yellow-necked mouse (*A. flavicollis*, $N=905$)															
Estimate	−2.481	−0.976	−0.797	0.055	1.03					1.099	2.522	0.594	1.423	−0.505	−1.929	
SE	0.329	0.284	0.346	0.015	0.46					0.395	0.389	0.4	0.33	0.348	0.343	
z-Value	−7.549	−3.44	−2.301	3.769	2.241					2.779	6.487	1.49	4.278	−1.45	−5.623	
p	**<0.001**	**<0.001**	**0.021**	**<0.001**	**0.025**					**0.028**	**<0.001**	0.445	**<0.001**	0.466	**<0.001**	
Variation (σ)																0.061
SD																0.247
Wood mouse (*A. sylvaticus*, $N=102$)															
Estimate	−2.802			0.167												
SE	0.518			0.059												
z-Value	−5.413			2.824												
p	**<0.001**			**0.005**												
Variation (σ)																0.424
SD																0.651

(*continued*)

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TABLE 4. (CONTINUED)

	Source of variation (fixed effects)														*Random*
Species	*Intercept*	*Sex (m)*	*Reproduction (1)*	*Weight*	*Sex (m): reproduction (1)*	*Sex (m): weight (g)*	*Spring vs. autumn*	*Summer vs. autumn*	*Summer vs. spring*	*MWP vs. BW*	*NRW vs. BW*	*TH vs. BW*	*NRW vs. MWP*	*TH vs. MWP*	*TH vs. NRW*	*State:site*
Bank vole (*M. glareolus*, $N=1{,}537$)																
Estimate	−3.869	0.364		0.01		0.107				0.898	2.34	0.813	1.433	−0.085	−1.528	
SE	0.307	0.201		0.03		0.046				0.385	0.318	0.348	0.297	0.33	0.249	
z-Value	−12.587	1.812		0.318		2.351				2.33	7.35	2.339	4.861	−0.256	−6.125	
p	**<0.001**	0.069		0.75		**0.019**				0.087	**<0.001**	0.086	**<0.001**	0.994	**<0.001**	
Variation (σ)																0.000
SD																0.000
Common shrew (*S. araneus*, $N=492$)																
Estimate	−2.148			0.357			0.965	0.353	−0.612							
SE	0.277			0.098			0.338	0.324	0.355							
z-Value	−7.752			3.656			2.854	1.089	−1.724							
p	**<0.001**			**<0.001**			**0.012**	0.52	0.196							
Variation (σ)																0.249
SD																0.499
Crowned shrew (*S. coronatus*, $N=101$)																
Estimate	−3.825															
SE	1.169															
z-Value	−3.271															
p	**0.001**															
Variation (σ)																6.538
SD																2.557
Eurasian pygmy shrew (*S. minutus*, $N=73$)																
Estimate	−2.607															
SE	0.527															
z-Value	−4.945															
p	**<0.001**															
Variation (σ)																0.726
SD																0.852

The nested spatial design was accounted for a random factor with a significance level of $\alpha=0.05$.
1, Reproduction active; m, male; SD, standard deviation; SE, standard error; state:site, site nested in state.
Siginificant values are highlighted in *bold*.

TABLE 5. INDIVIDUAL INFECTION OF COMMON, FIELD, AND BANK VOLE, YELLOW-NECKED, WOOD, AND STRIPED FIELD MOUSE, COMMON, CROWNED, AND EURASIAN PYGMY SHREW WITH *LEPTOSPIRA* SPP. ON FOREST VERSUS GRASSLAND PER SEASON

Source of variation			*Estimate*	*SE*	z	p	*Variance*	*SD*
Spring	Intercept		−1.154	0.514	2.244	0.025		
	Forest vs. grassland		−0.909	0.434	−2.096	0.036		
	Random	Fed					0.396	0.630
		Fed:site					0.515	0.718
Summer	Intercept		−1.265	0.330	−3.833	<0.001		
	Forest vs. grassland		−1.247	0.275	−4.527	<0.001		
	Random	Fed					0.169	0.411
		Fed:site					0.251	0.501
Autumn	Intercept		−1.326	0.251	−5.278	<0.001		
	Forest vs. grassland		−0.977	0.325	−3.005	0.002		
	Random	Fed					0.384	0.619
		Fed:site					0.000	0.000

The model incorporates all species and the nested spatial design as a random factor with a significance level of $\alpha = 0.05$.
Fed, federal state; fed:site, site nested in federal state.

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for *L. interrogans* ST 24, which was found in rodents in forest habitats and *L. borgpetersenii* ST 146 detected in common and crowned shrews. *L. interrogans* ST 24 was also found in beavers from South West Germany (Woll et al. 2012), which also suggests an infection of the beavers by contact with forest-dwelling rodents or rodent-contaminated water.

In a previous study, *L. interrogans* and *L. borgpetersenii* were detected in field voles (Mayer-Scholl et al. 2014), questioning a simple ''one pathogen-one carrier'' pattern. Field voles mainly inhabit the contact zone between forest and grassland and can shift between both habitats (Borowski 2003, Fey et al. 2006). An interesting question is whether the field vole is involved in the transmission of *Leptospira* spp. between grassland and forest habitats, particularly for *L. kirschneri* ST 110.

Influences of demographic factors and seasonal fluctuations on Leptospira *prevalence*

The role of common and field voles as the main hosts of *L. kirschneri* is also supported by the positive correlation of weight (age) and *Leptospira* prevalence. This was observed for most of the studied species, suggesting a persisting infection in the whole rodent and shrew community with prevalence related to time (age) rather than population abundance (Faine 1999, Levett 2001, Krojgaard et al. 2009, Perez et al. 2011, Benacer et al. 2013, Himsworth et al. 2013). However, it cannot be excluded that the correlation of weight (age) and *Leptospira* prevalence is caused by an age-dependent increase of the susceptibility to infection. Other demographic factors were not consistently associated with *Leptospira* infection across species. For example, in accordance with Mayer-Scholl et al. (2014), sex was not associated with *Leptospira* prevalence in *Microtus* species, which suggests that sex-specific inter-individual social contact such as fighting for territory and mates may not be as important as time (age) of exposure. This is in contrast to previous studies on urban rats, which detected an influence of sex on prevalence (Easterbrook et al. 2007, Benacer et al. 2013). Such an influence in some urban areas in other regions of the world, besides Europe, may indicate a stronger man-influenced effect on the respective rat populations in the investigated areas, as the majority of published studies did not report an influence of sex on *Leptospira* prevalence (Collares-Pereira et al. 2000, Agudelo-Florez et al. 2009, Halliday et al. 2013, Costa et al. 2014, Mayer-Scholl et al. 2014, Heuser et al. 2017).

Implications for human infections

Finally, a prevalence of 30% in common voles suggests the occurrence of several hundred infected individuals per hectare during vole outbreaks (Bryja et al. 2005, Cornulier et al. 2013). *L. kirschneri* ST 110 found in our and previous studies in common and field voles corresponds to the serovar Grippotyphosa, a serovar that was previously observed as causative for leptospirosis outbreaks in strawberry pickers in Germany (Desai et al. 2009, Dreesman et al. 2016). For these outbreaks, common voles were identified as the most likely source of human infections (Desai et al. 2009). In comparison to the years 2010–2013, the increased *Leptospira* prevalence in common and field voles during 2014 (Supplementary Fig. S3A, B) may explain the disease cluster in strawberry pickers in Lower Saxony in 2014. The potential involvement of larger animals in leptospirosis epidemiology could not be evaluated in our study, as only a single report from North Rhine-Westphalia detected *Leptospira* prevalence of 26% in wild boar (Schönberg et al. 1999).

Our study showed the highest peak in *Leptospira* prevalence in animals, respectively, common and field voles, in autumn. However, this does not seem to correlate with the increased incidence of human leptospirosis cases during summer reported by the Robert Koch-Institute in Germany (Robert Koch-Institute 2016). An increase of the number of recorded human leptospirosis cases during the summer months is most likely due to the increased human exposure through outdoor activities in agriculture and forestry, and for leisure. The observed discrepancies between *Leptospira* prevalence in common and field voles and the reported overall low number of human infections might be explained by an underreporting of human leptospirosis cases due to nonspecific and variable symptoms of the disease. The high seroprevalence of 14% in forestry workers and 4% in the general population from a randomized population-based

cross-sectional study in Germany (Jurke et al. 2015, Brockmann et al. 2016) indicates that the *Leptospira* exposure and seroconversion are much more frequent than commonly assumed. Therefore, future studies are required to evaluate the occurrence of human *Leptospira* infections and potential associations with its reservoir species, as well as factors regarding human exposure.

Conclusion

The study confirmed a broad geographical distribution of *Leptospira* across rodent and shrew species and underlines the relevance of common and field voles as potential reservoirs of *L. kirschneri* ST 110. The sharing of STs among different rodent and shrew species occurring in the same habitats and the emergence of a novel *L. kirschneri* ST 230 in common voles in Baden-Wuerttemberg highlight the need for further investigations by means of molecular typing to better understand the epidemiology of leptospires and their impact on public health. Further investigations should focus on population dynamics of common and field vole and their impact on *Leptospira* prevalence and its putative influence on the human infection risk. In addition, comparative studies are needed to evaluate if *L. interrogans* STs in commensal rats and forest rodents have an impact on public health in rural and urban regions, at least in outbreak scenarios.

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